Kevin Wang

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EDUCATION

University of California, Santa Cruz — B.S. Electrical Engineering | Expected 2027 | GPA: 3.42

Relevant Coursework: Analog Electronics, High-Speed Digital Design, Energy Conversion & Control, Signals & Systems, Logic Design, Electronic Design Automation, Feedback Control Systems

EXPERIENCE

Human-Aware Robotic Exploration (HARE) Lab, UCSC — Undergraduate Student Researcher Sep 2026 – Present

- Developing **visual-inertial SLAM** localization for agricultural UAVs where **RTK GNSS** is unreliable.
- Evaluating **ORB-SLAM3** and **Structure-from-Motion (SfM)** for pose estimation and 3D mapping.
- Integrating **camera** and **IMU** sensors with onboard systems, focusing on timing and calibration.
- Connecting **SLAM** localization to the UAV's **ROS 2** stack for flight and mapping.

Formula Slug (UCSC FSAE) — Low Voltage Electrical Shutdown Engineer Sep 2024 – Jun 2025

- Worked on the design and testing of shutdown circuitry for the vehicle's tractive system.
- Tested proposed circuit designs in **LTspice** to check logic, timing, and behavior under fault conditions.
- Analyzed **BMS** and **IMD** fault logic, relay control, and supporting analog circuits using schematics and datasheets.
- Worked with electrical and mechanical team members to make sure shutdown requirements were properly integrated into the vehicle.

Digsur Tech — Outreach & Communications Intern Sep 2025 – Dec 2025

- Presented drone-based surveying and 3D rendering services to prospective clients.
- Explained drone capabilities, applications, and operating use cases during customer presentations.
- Gained experience with commercial drone systems used for surveying, geospatial data collection, and 3D environment reconstruction.

Engineering & Physics Tutor Oct 2025 – Present

- Tutored circuit labs involving hardware wiring, schematic reading, measurements, and debugging.
- Taught the behavior and use of **op-amps**, **diodes**, **potentiometers**, **DACs**, **flip-flops**, and **inverter ICs**.
- Helped students use **multimeters**, **oscilloscopes**, **spectrum analyzers**, and **AC/DC power supplies** during lab work.
- Tutored signals and systems, differential equations, electrostatics, and classical mechanics.

PROJECTS

High-Power DC Electronic Load Test Bench In Development

- Developing a portable electronic load for DC power-source testing with a target range up to 150 W, 30 V, and 10 A.
- Designing the PCB in **KiCad** around three parallel power **MOSFETs**, current-sense circuitry, op-amp feedback, protection, and low-voltage support electronics.
- Planning forced-air heatsinking, enclosure-mounted controls, and a compact assembly for repeatable bench and portable testing.
- Designing an adaptable connector interface for USB-C, USB, barrel-jack, and other DC source connections.

Programmable Electronic Load and Test Platform

- Designed and built a programmable constant-current load for testing DC power sources at different load currents.
- Used an **LM324 op-amp**, **MOSFET**, sense resistor, and **DAC** reference to create closed-loop current control.
- Connected an **ESP32** microcontroller, **MCP4725 DAC**, **OLED display**, and rotary encoder for current adjustment and system feedback.
- Wrote embedded **C** firmware for **I²C** communication, **DAC** control, display output, and rotary encoder input.
- Traced **I²C** and hardware problems through voltage measurements, continuity checks, clock-rate changes, and grounding tests, finding a missing ground trace on the **DAC** module from a manufacturing defect.

High-Speed Digital Circuit Demonstration PCB

- Designed and assembled a 2-layer **PCB** to demonstrate ground bounce, channel loss, **NEXT**, and **FEXT**.
- Used return paths, parasitic inductance, coupling, and switching behavior to guide the circuit and **PCB** design.
- Created test structures with different trace spacing, routing lengths, and geometries to examine individual signal-integrity effects.
- Used an **oscilloscope** and **spectrum analyzer** to measure waveform distortion and frequency-domain behavior.
- Added **SMA** connectors and dedicated test points to make high-frequency measurements easier to repeat.

Audio Amplifier PCB

- Built and tested a fabricated two-layer analog audio amplifier **PCB**.
- Turned schematics, datasheets, and component requirements into a manufacturable board using **OrCAD** and **Allegro**.
- Created custom footprints from package dimensions, pad geometry, drill sizes, and component spacing.
- Routed analog signal and power connections with attention to grounding, current paths, and noise-sensitive circuitry.

Basys3 VGA Game

- Built a simplified Subway Surfers-style game on a **Basys 3 FPGA** using behavioral **Verilog** and state-machine logic.
- Created 60 FPS **VGA** output with pixel-level graphics for player movement, tracks, and multi-color objects.
- Generated randomized train obstacles across three tracks with different lengths, positions, and spawn intervals.
- Added pixel-based collision detection, jump timing, reset control, and an optional invincibility mode.

SKILLS

PCB / EDA: OrCAD Capture, Allegro PCB Designer, PSpice/LTspice, Altium Designer, KiCad, GerbTool

Programming / Simulation: Verilog, Python, C, MATLAB, Simulink, ROS 2, ANSYS Maxwell, dSPACE, Data Parsing

Hardware / Testing: Oscilloscopes, Spectrum Analyzers, Function Generators, Multimeters, AC/DC Power Supplies, Breadboarding, Signal Tracing

Electrical Engineering: Analog Electronics, Op-Amp & MOSFET Circuits, Feedback & Current Regulation, Digital Logic & State Machines, Signals & Systems, Signal Integrity, High-Speed Digital Design, Power & Energy Conversion, I²C Interfaces, PCB Design, Circuit Simulation, System Validation, Hardware Bring-Up & Debugging